SUPPORTING ADULT GUIDANCE

Perimeter or area?

Perimeter and Area · Lesson 1. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

In the main activity, Fence and Field, learners drag the corner of a rectangular plot on a grid of 1 m squares; the fence (perimeter) and the grass (area) are counted live, with a post every metre and every square drawn. Two fixed modes follow: with the fence fixed at 20 m, learners find all five rectangles (areas 9 to 25 m^2); with the grass fixed at 12 m^2, all three (fences 14 to 26 m). A table records what they find. Six questions follow (answers with the wrong unit are caught), and a stretch shows why the square holds the most grass. Learners then sort real jobs into perimeter or area, judge statements, and design plots to a brief.

Driving Question: *Can two shapes have the same perimeter but different areas?*

The mathematics, in plain terms

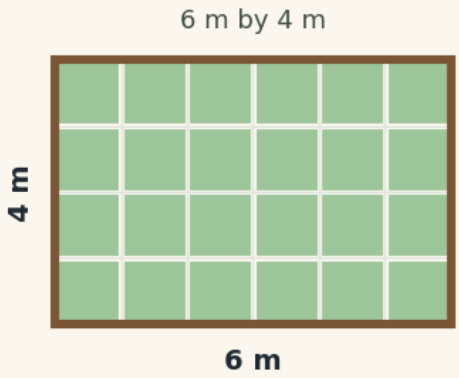
Perimeter is a length (m); area is a measure of surface (m^2). For rectangles with a fixed perimeter, the area is greatest for the square; for a fixed area, the perimeter is least for the square. The stretch shows this concretely: turning 5 × 5 into 4 × 6 removes a column of 5 and adds a row of 4, losing one square. All lengths are whole metres, so every number shown is exact. Area by multiplication (length × width) is formalised in Lesson 2.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Fence and grass

Find the perimeter and the area of this plot.

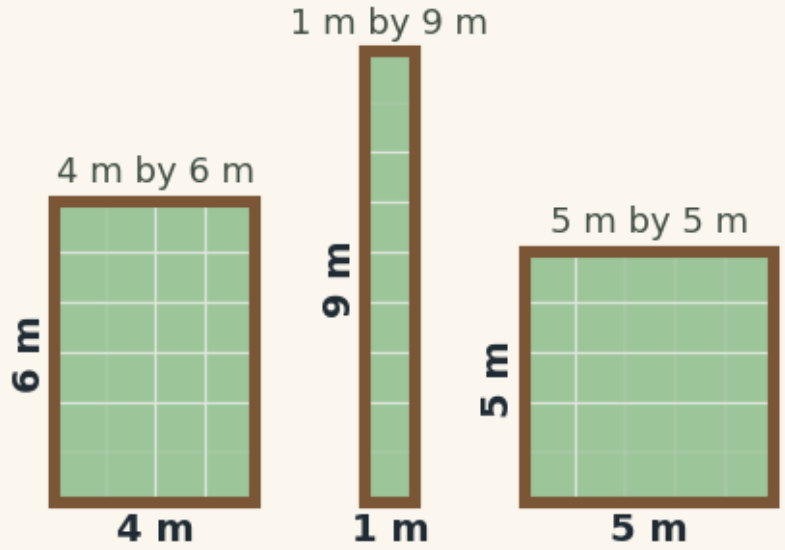


1. Perimeter: add all four sides.	6 + 4 + 6 + 4 = 20 m
2. Area: count the squares inside.	4 rows of 6 = 24 squares
3. Each square is 1 m^2, so	the area is 24 m^2

Perimeter 20 m; area 24 m^2.

Example 2 — Same fence, different grass

Each plot uses 20 m of fence. How much grass does each hold?



1. 4 m by 6 m: 4 + 6 + 4 + 6 = 20 m of fence. 24 m^2 of grass
2. 1 m by 9 m: 1 + 9 + 1 + 9 = 20 m of fence. 9 m^2 of grass
3. 5 m by 5 m: 5 + 5 + 5 + 5 = 20 m of fence. 25 m^2 of grass

Same perimeter, different areas. The square holds the most.

Feel the difference, then fix one and watch the other

Learners see the fence and the grass as different things on the same picture before either is named formally. Fixing one at a time makes the independence of perimeter and area unmistakable. The sorting and the design briefs then use the distinction.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Fence and Field, the fixed-fence and fixed-grass hunts, and six questions. Stretch: why the square holds the most.	~16 min
Movement	Perimeter or area? Sorting real jobs.	~7 min
Reflection	True or not? Perimeter and area mix-ups.	~7 min
Creativity	Design a plot to a brief.	~10 min

Rest	Same fence, different fields.	~2 min
Nutrition	Intellectual: careful questions.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Thinking a bigger perimeter always means a bigger area.
Ask: “Compare 1 m by 9 m with 4 m by 4 m. Which has more fence? Which has more grass?”

Giving an area in m, or a perimeter in m^2.
Ask: “Is this a distance round, or squares inside?”

Adding only two sides for a perimeter.
Ask: “Walk your finger all the way round. How many sides did you pass?”

Counting the squares along the edge as the perimeter.
Ask: “Is the fence made of squares, or of metres of edge?”

Thinking area and perimeter are the same thing for a square.
Ask: “A 4 m square has 16 m of fence and 16 m^2 of grass. Are those the same kind of thing?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: counting squares, and metres (Place Value Lesson 8). A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner says a 3 m by 4 m rectangle has a perimeter of 12 m. The aim is a question that separates round from inside.

- “What did you do with the 3 and the 4?”
- “Is 12 the number of squares inside, or the distance round?”

- “Can you trace the edge and add the sides?”
- “Which unit goes with each number?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Why only whole metres?
So every number shown is exact and countable. Decimals and other shapes come in later lessons.

Should learners use length × width yet?
If they spot it, celebrate it; Lesson 2 builds it properly. Counting rows is enough here.

Is the stretch required?
No. The secure core is telling perimeter from area, finding both for a rectangle, and using the right units.

What comes next?
Lesson 2: area of rectangles and triangles.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.

Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.